

Reinforcement Learning

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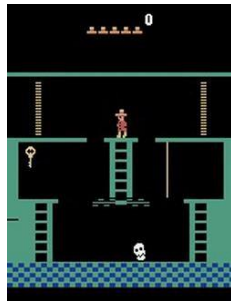
Transfer Learning and **Meta-Learning**

What's the problem?

this is easy (mostly)

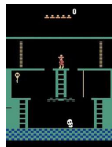
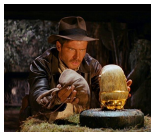


this is difficult



Why?

Can RL use the same prior knowledge as us?



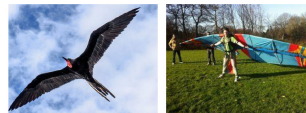
- ▶ If we've solved prior tasks, we might acquire useful knowledge for solving a new task
- ▶ How is the knowledge stored?
 - **Q-function:** tells us which actions or states are good
 - **Policy:** tells us which actions are potentially useful
 - ▶ some actions are never useful!
 - **Models:** what are the laws of physics that govern the world?
 - **Features/hidden states:** provide us with a good representation
 - ▶ Don't underestimate this!

How can we frame transfer learning problems?

1. **Forward transfer:** learn policies that transfer effectively
 - a) Train on source task, then run on target task (or finetune)
 - b) Relies on the tasks being quite similar!
2. **Multi-task transfer:** train on many tasks, transfer to a new task
 - a) Sharing representations and layers across tasks in multi-task learning
 - b) New task needs to be similar to the distribution of training tasks
3. **Meta-learning:** learn to learn on many tasks
 - a) Accounts for the fact that we'll be adapting to a new task during training!

What issues are we likely to face?

- ▶ **Domain shift:** representations learned in the source domain might not work well in the target domain
- ▶ **Difference in the MDP:** some things that are possible to do in the source domain are not possible to do in the target domain
- ▶ **Finetuning issues:** if pretraining & finetuning, the finetuning process may still need to explore, but optimal policy during finetuning may be deterministic!

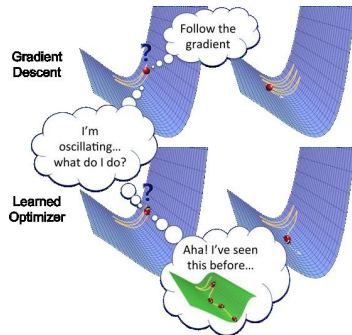


How can we frame transfer learning problems?

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What is meta-learning?

- ▶ If you've learned 100 tasks already, can you figure out how to learn more efficiently?
 - Now having multiple tasks is a huge advantage!
- ▶ Meta-learning = learning to learn
- ▶ In practice, very closely related to multi-task learning
- ▶ Many formulations
 - Learning an optimizer
 - Learning an RNN that ingests experience
 - Learning a representation



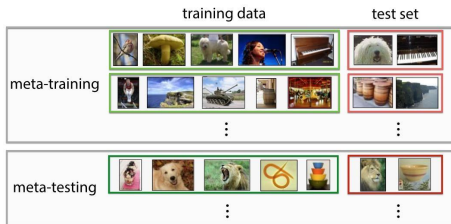
Why is meta-learning a good idea?

- ▶ Deep reinforcement learning, especially model-free, requires a huge number of samples
- ▶ If we can meta-learn a faster reinforcement learner, we can learn new tasks efficiently!
- ▶ What can a meta-learned learner do differently?
 - Explore more intelligently
 - Avoid trying actions that are known to be useless
 - Acquire the right features more quickly

Meta-learning with supervised learning



image credit: Ravi & Larochelle '17

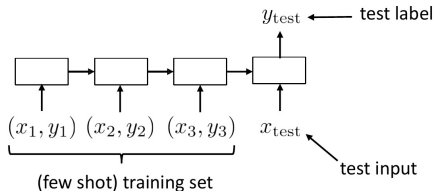


supervised learning: $f(x) \rightarrow y$

input (e.g., image) output (e.g., label)

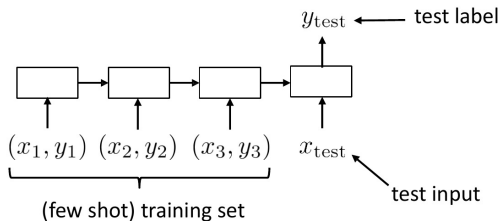
supervised meta-learning: $f(\mathcal{D}^{\text{tr}}, x) \rightarrow y$

training set



- How to read in training set?
- Many options, RNNs can work
 - More on this later

What is being “learned” ?



supervised meta-learning:

$$f(\mathcal{D}^{\text{tr}}, x) \rightarrow y$$

“Generic” learning:

$$\begin{aligned}\theta^* &= \arg \min_{\theta} \mathcal{L}(\theta, \mathcal{D}^{\text{tr}}) \\ &= f_{\text{learn}}(\mathcal{D}^{\text{tr}})\end{aligned}$$

“Generic” meta-learning:

$$\theta^* = \arg \min_{\theta} \sum_{i=1}^n \mathcal{L}(\phi_i, \mathcal{D}_i^{\text{ts}})$$

where $\phi_i = f_{\theta}(\mathcal{D}_i^{\text{tr}})$

What is being “learned” ?

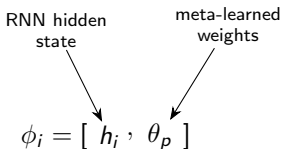
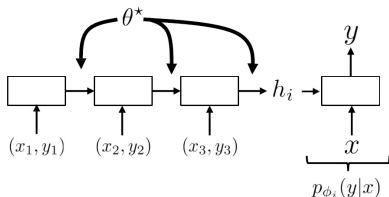
“Generic” learning:

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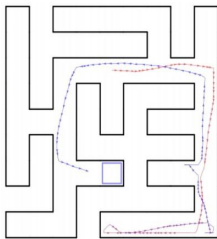
where $\phi_i = f_{\theta}(\mathcal{D}_i^{\text{tr}})$



Meta Reinforcement Learning

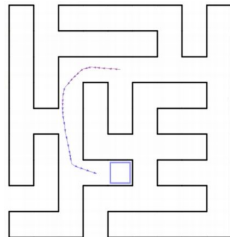
Goal:

Collect small amount of experience in new
MDP



Collect $\mathcal{D}_{\text{tr}} \sim \pi^{\text{exp}}$

Learn policy that solves that MDP

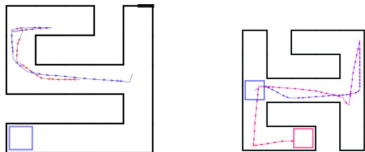


$\mathcal{D}_{\text{tr}} \rightarrow \pi^{\text{task}}$

diagram adapted from Duan et al. '17

Meta-Train Time:

Learn how to efficiently explore & solve many MDPs:

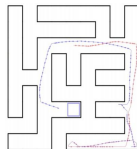


... meta-training tasks

Meta-train $\pi^{\text{exp}}, \pi^{\text{task}}$

Meta-Test Time:

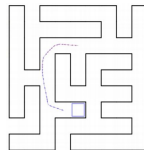
Collect small amount of experience in new MDP



Collect $\mathcal{D}_{\text{tr}} \sim \pi^{\text{exp}}$

Learn policy that solves that

MDP



$\mathcal{D}_{\text{tr}} \rightarrow \pi^{\text{task}}$

Key assumption: Meta-training & meta-testing MDPs come from the same distribution (so that we can expect generalization).

diagram adapted from Duan et al. '17

“Generic” learning:

$$\begin{aligned}\theta^* &= \arg \min_{\theta} \mathcal{L}(\theta, \mathcal{D}^{\text{tr}}) \\ &= f_{\text{learn}}(\mathcal{D}^{\text{tr}})\end{aligned}$$

Reinforcement learning:

$$\begin{aligned}\theta^* &= \arg \max_{\theta} \mathbb{E}_{\pi_{\theta}}[R(\tau)] \\ &= f_{\text{RL}}(\mathcal{M}) \quad \mathcal{M} = \{\mathcal{S}, \mathcal{A}, \mathcal{P}, r\}\end{aligned}$$

MDP



“Generic” meta-learning:

$$\theta^* = \arg \min_{\theta} \sum_{i=1}^n \mathcal{L}(\phi_i, \mathcal{D}_i^{\text{ts}})$$

where $\phi_i = f_{\theta}(\mathcal{D}_i^{\text{tr}})$

Meta-reinforcement learning:

$$\theta^* = \arg \max_{\theta} \sum_i \mathbb{E}_{\pi_{\phi_i}}[R(\tau)]$$

where $\phi_i = f_{\theta}(\mathcal{M}_i)$

MDP for task i



The meta reinforcement learning problem

$$\theta^* = \arg \max_{\theta} \sum_i \mathbb{E}_{\pi_{\phi_i}} [R(\tau)], \quad \phi_i = f_{\theta}(\mathcal{M}_i)$$

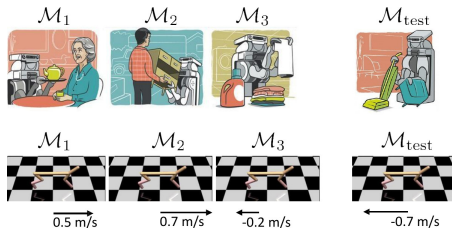
assumption: $\mathcal{M}_i \sim p(\mathcal{M})$

meta test-time: sample $\mathcal{M}_{\text{test}} \sim p(\mathcal{M})$, get
 $\phi_i = f_{\theta}(\mathcal{M}_{\text{test}})$

$\{\mathcal{M}_1, \dots, \mathcal{M}_n\}$

meta-training MDPs

Some examples:



$$\theta^* = \arg \max_{\theta} \sum_{i=1}^n \mathbb{E}_{\pi_{\phi_i}(\tau)} [R(\tau)] \quad \Rightarrow \quad \theta^* = \arg \max_{\theta} \sum_{i=1}^n \mathbb{E}_{\pi_{\theta}} [R(\tau)]$$

where $\phi_i = f_{\theta}(\mathcal{M}_i)$

$\pi_{\theta}(a_t \mid \underbrace{s_t, s_1, a_1, r_1, \dots, s_{t-1}, a_{t-1}, r_{t-1}}_{\text{context used to infer whatever we need to solve } \mathcal{M}_i, \text{ i.e., } z_t \text{ or } \phi_i \text{ (which are really the same thing)}}$)

in meta-RL, the *context* is inferred from experience from $\mathcal{M}_i$

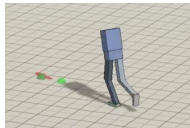
in multi-task RL, the context is typically given

$$\pi_{\theta}(a_t \mid s_t, \phi_i)$$

“context”



ϕ : stack location



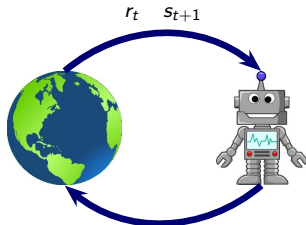
ϕ : walking direction



ϕ : where to hit puck

$$\theta^* = \arg \max_{\theta} \sum_{i=1}^n \mathbb{E}_{\pi_{\phi_i}(\tau)} [R(\tau)]$$

where $\phi_i = f_{\theta}(\mathcal{M}_i)$



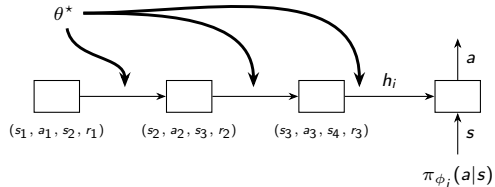
pick $a_t \sim \pi_{\theta}(a_t | s_t)$

use (s_t, a_t, s_{t+1}, r_t) to improve π_{θ}

main question: how to implement $f_{\theta}(\mathcal{M}_i)$? what should $f_{\theta}(\mathcal{M}_i)$ do?

1. improve policy with experience from $\mathcal{M}_i$:
 $\{(s_1, a_1, s_2, r_1), \dots, (s_T, a_T, s_{T+1}, r_T)\}$

2. (new in RL): choose how to interact, i.e. choose a_t
 — meta-RL must also *choose* how to *explore*!

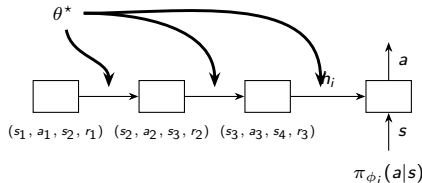


as before, $\phi_i = [h_i, \theta_{\pi}]$

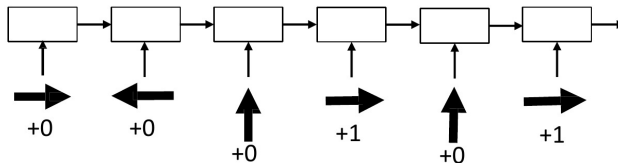
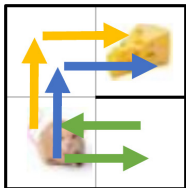
$$\theta^* = \arg \max_{\theta} \sum_{i=1}^n \mathbb{E}_{\pi_{\phi_i}(\tau)} [R(\tau)]$$

where $\phi_i = f_{\theta}(\mathcal{M}_i)$

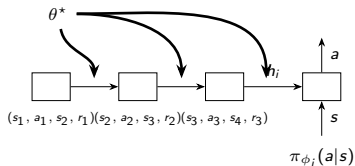
so... we just train an RNN
policy?
yes!



crucially, RNN hidden state is **not** reset between episodes!

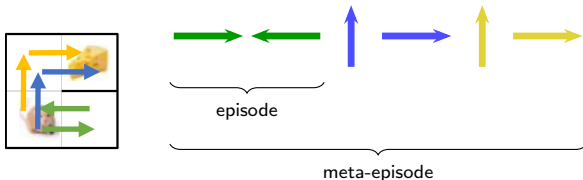


Why recurrent policies learn to explore



1. improve policy with experience from $\mathcal{M}_i$: $\{(s_1, a_1, s_2, r_1), \dots, (s_T, a_T, s_{T+1}, r_T)\}$
2. (new in RL): choose how to interact, i.e. choose a_t
meta-RL must also *choose* how to *explore*!

$$\theta^* = \arg \max_{\theta} \mathbb{E}_{\pi_{\theta}} \left[\sum_{t=0}^T r(s_t, a_t) \right]$$



optimizing total reward over the entire **meta-episode** with RNN policy **automatically** learns to explore!

Meta-Training

1. Sample task $\mathcal{T}_i$
2. Roll-out policy $\pi(a \mid s, \mathcal{D}_i^{\text{tr}})$ for N episodes (under dynamics $p_i(s' \mid s, a)$ and reward $r_i(s, a)$)
3. Store sequence in replay buffer for task $\mathcal{T}_i$
4. Update policy to maximize discounted return for all tasks.
5. **Repeat** (go back to step 1).

Meta-Test Time

1. Sample new task $\mathcal{T}_j$
2. Roll-out policy $\pi(a \mid s, \mathcal{D}_j^{\text{tr}})$ for up to N episodes

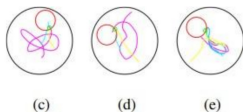
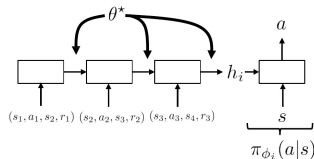
Experiment: Learning to visually navigate a maze — train on 1000 small mazes, test on held-out small mazes and large mazes

Method	Small Maze		Large Maze	
	Episode 1	Episode 2	Episode 1	Episode 2
Random	188.6 ± 3.5	187.7 ± 3.5	420.2 ± 1.2	420.8 ± 1.2
LSTM	52.4 ± 1.3	39.1 ± 0.9	180.1 ± 6.0	150.6 ± 5.9
SNAIL (ours)	50.3 ± 0.3	34.8 ± 0.2	140.5 ± 4.2	105.9 ± 2.4

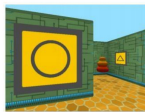
From: Mishra, Rohaninejad, Chen, Abbeel. A Simple Neural Attentive Meta-Learner. ICLR 2018

$$\theta^* = \arg \max_{\theta} \sum_{i=1}^n E_{\pi_{\phi_i}(\tau)} [R(\tau)]$$

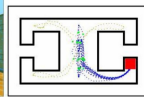
where $\phi_i = f_{\theta}(\mathcal{M}_i)$



Heess. Hunt. Lillicrap. Silver. **Memory-based control with**

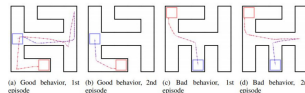


(a) Labyrinth maze



(b) Illustrative Episode

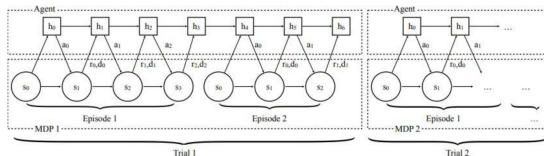
Wang. Kurth-Nelson. Tirumala. Sover. Leibo. Munos.



(a) Good behavior, 1st episode (b) Good behavior, 2nd episode (c) Bad behavior, 1st episode (d) Bad behavior, 2nd episode

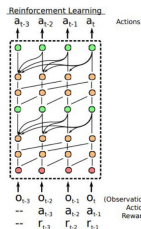
Duan. Schulman. Chen. Bartlett. Sutskever. Abbeel. **RL²:**

Heess et al. Memory-based control with recurrent neural networks. 2015. • Wang et al. Learning to Reinforcement Learn. 2016. • Duan, Schulman, Chen, Bartlett, Sutskever, Abbeel. **RL²: Fast Reinforcement Learning via Slow Reinforcement Learning.** 2016.



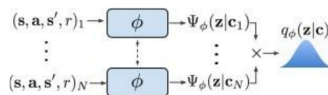
standard RNN (LSTM) architecture

Duan, Schulman, Chen, Bartlett, Sutskever, Abbeel. **RL2: Fast Reinforcement Learning via Slow Reinforcement Learning**. 2016.



attention + temporal convolution

Mishra, Rohaninejad, Chen, Abbeel. **A Simple Neural Attentive Meta-Learner**.



parallel permutation-invariant context encoder

Rakelly*, Zhou*, Quillen, Finn, Levine. **Efficient Off-Policy Meta-Reinforcement learning via Probabilistic Context Variables**.

Challenges and **Open Problems** in RL

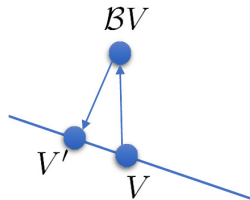
Challenges with core algorithms:

- ▶ **Stability:** does your algorithm converge?
- ▶ **Efficiency:** how long does it take to converge? (how many samples)
- ▶ **Generalization:** after it converges, does it generalize?

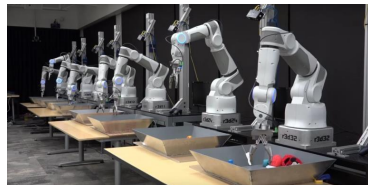
Challenges with assumptions:

- ▶ Is this even the right problem formulation?
- ▶ What is the source of supervision?

- ▶ Devising stable RL algorithms is very hard
- ▶ Q-learning / value function estimation
 - Fitted Q / fitted value methods with deep network function estimators are typically not contractions, hence no guarantee of convergence
 - Lots of parameters for stability: target network delay, replay buffer size, clipping, sensitivity to learning rates, etc.
- ▶ Policy gradient / likelihood ratio / REINFORCE
 - Very high variance gradient estimator
 - Lots of samples, complex baselines, etc.
 - Parameters: batch size, learning rate, design of baseline
- ▶ Model-based RL algorithms
 - Model class and fitting method
 - Optimizing policy w.r.t. model non-trivial due to backpropagation through time
 - More subtle issue: policy tends to exploit the model

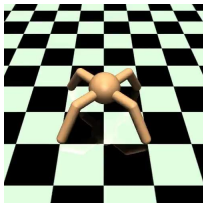


- ▶ Need to wait for a long time for your homework to finish running
- ▶ Real-world learning becomes difficult or impractical
- ▶ Precludes the use of expensive, high-fidelity simulators
- ▶ Limits applicability to real-world problems





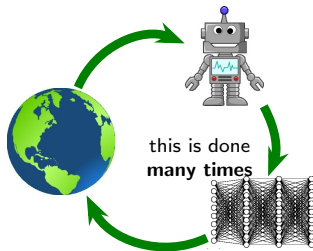
- ▶ Large-scale
- ▶ Emphasizes diversity
- ▶ Evaluated on generalization



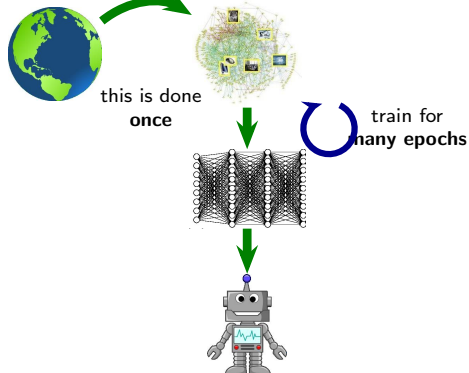
- ▶ Small-scale
- ▶ Emphasizes mastery
- ▶ Evaluated on performance
- ▶ Where is the generalization?

RL has a big problem

reinforcement learning

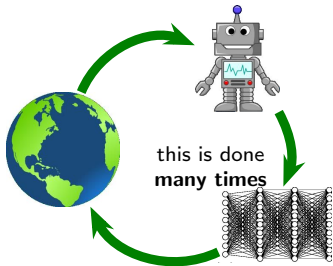


supervised machine learning

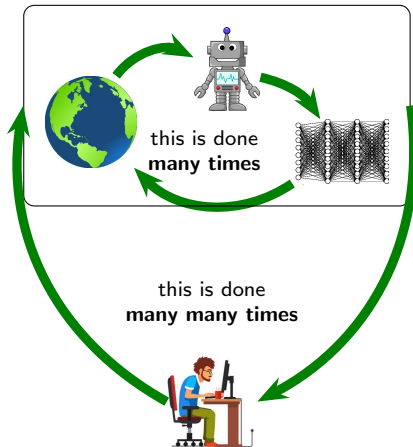


RL has a big problem

reinforcement learning

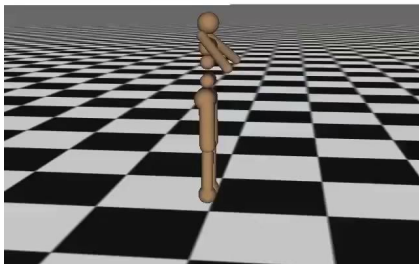


actual reinforcement learning



How bad is it?

Iteration 0



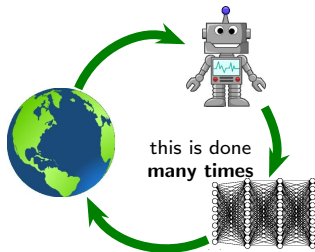
- ▶ This is quite cool
- ▶ It takes 6 days of real time (if it was real time)
- ▶ ...to run on an infinite flat plane



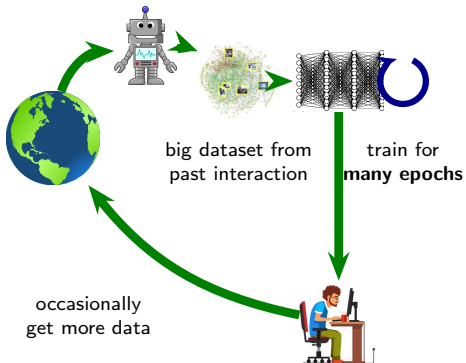
The real world is not so simple!

Schulman, Moritz, L., Jordan, Abbeel '16

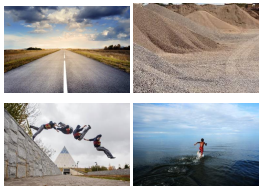
reinforcement learning



off-policy reinforcement learning



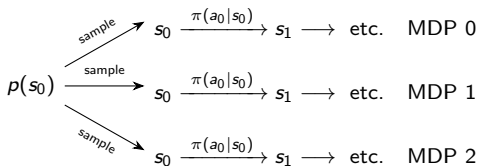
Single task or multi-task?



- ▶ this is where generalization can come from. . .
- ▶ maybe doesn't require any new assumption, but might merit additional treatment

The real world is not so simple!

pick MDP randomly
in first state



Where does the supervision come from?

- ▶ If you want to learn from many different tasks, you need to get those tasks somewhere!
- ▶ Learn objectives / rewards from demonstration (inverse reinforcement learning)
- ▶ Generate objectives automatically?



Mnih et al. '15

reinforcement learning agent



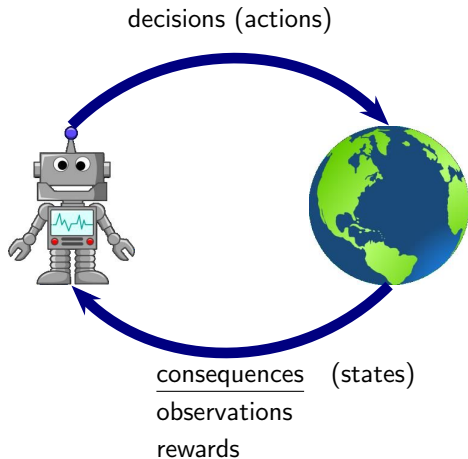
what is the **reward**?

- ▶ How should we define a control problem?
 - What is the data?
 - What is the goal?
 - What is the supervision? (may not be the same as the goal...)
- ▶ Think about the assumptions that fit your problem setting!
- ▶ Don't assume that the basic RL problem is set in stone

- ▶ Reinforcement Learning as an Engineering Tool
- ▶ Reinforcement Learning and the Real World
- ▶ Reinforcement Learning as “Universal” Learning

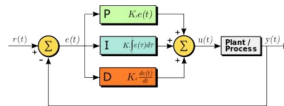
Reinforcement Learning as an **Engineering Tool**

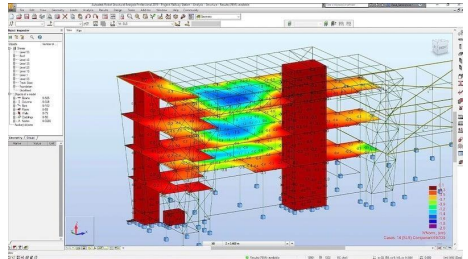
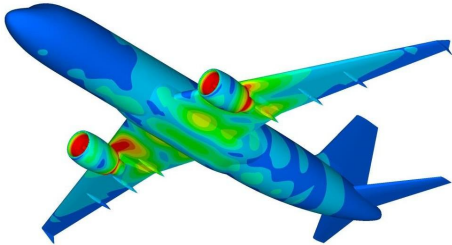
What we think RL is...





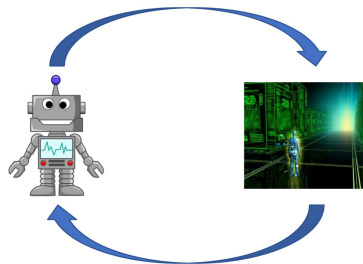
$$\begin{aligned} \mathbf{r} &= \mathbf{r}(t) = r \hat{\mathbf{e}}_r \\ \mathbf{v} &= v \hat{\mathbf{e}}_r + r \frac{d\theta}{dt} \hat{\mathbf{e}}_\theta + r \frac{d\varphi}{dt} \sin \theta \hat{\mathbf{e}}_\varphi \\ \mathbf{a} &= \left(a - r \left(\frac{d\theta}{dt} \right)^2 - r \left(\frac{d\varphi}{dt} \right)^2 \sin^2 \theta \right) \hat{\mathbf{e}}_r \\ &\quad + \left(r \frac{d^2\theta}{dt^2} + 2v \frac{d\theta}{dt} - r \left(\frac{d\varphi}{dt} \right)^2 \sin \theta \cos \theta \right) \hat{\mathbf{e}}_\theta \\ &\quad + \left(r \frac{d^2\varphi}{dt^2} \sin \theta + 2v \frac{d\varphi}{dt} \sin \theta + 2r \frac{d\theta}{dt} \frac{d\varphi}{dt} \cos \theta \right) \hat{\mathbf{e}}_\varphi \end{aligned}$$





RL: anything you can simulate you can control

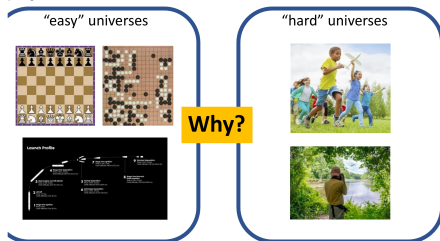
- ▶ Provides a powerful engineering tool
- ▶ Not that different from conventional engineering approach!
 - Before: characterize, simulate, control
 - Now: characterize, simulate, run RL
- ▶ Main role: powerful inversion engine
- ▶ Main weakness: still need to simulate!



Reinforcement Learning and the **Real World**



Moravec's paradox seems like a statement about AI but it is actually a statement about the physical universe.



"We are all prodigious olympians in perceptual and motor areas, so good that we make the difficult look easy. Abstract thought, though, is a new trick, perhaps less than 100 thousand years old. We have not yet mastered it. It is not all that intrinsically difficult; it just seems so when we do it."
– Hans Moravec

"The main lesson of thirty-five years of AI research is that the hard problems are easy and the easy problems are hard. The mental abilities of a four-year-old that we take for granted – recognizing a face, lifting a pencil, walking across a room, answering a question – in fact solve some of the hardest engineering problems ever conceived."
– Steven Pinker

What does this have to do with RL?



How do we engineer a system that can deal with the unexpected?

- ▶ Minimal external supervision about what to do
 - ▶ Unexpected situations that require adaptation
 - ▶ Must discover solutions autonomously
 - ▶ Must “stay alive” long enough to discover them!
-
- ▶ Humans are extremely good at this
 - ▶ Current AI systems are extremely bad at this
 - ▶ RL in principle can do this, and nothing else can

So what's the problem?

RL in principle can do this, and nothing else can — *but we rarely study this kind of setting in RL research!*

“easy” universes	“hard” universes
success = high reward (“optimal control”)	success = “survival” (“good enough control”)
closed world, rules are known	open world, everything must come from data
lots of simulation	no simulation (because rules are unknown)
Main question: can RL algorithms optimize really well	Main question: can RL generalize and adapt

RL should be really good in the “hard” universes!

Some questions that come up in the real world

- ▶ How do we tell RL agents what we want them to do?
- ▶ How can we learn fully autonomously in continual environments?
- ▶ How do remain robust as the environment changes around us?
- ▶ What is the right way to generalize using experience & prior data?
- ▶ What is the right way to bootstrap exploration with prior experience?



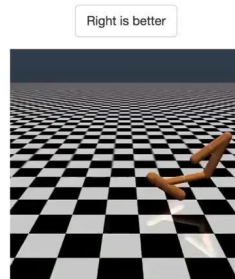
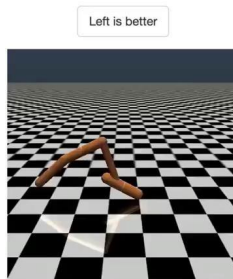
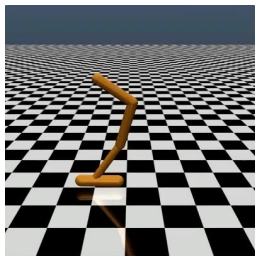
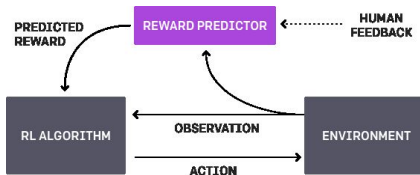
This is not about robots

Robots are the most natural for us to think about, because they are embodied like we are



This is not about robots — robots are just the most natural for us to think about, because they are embodied like we are

Other ways to communicate objectives?

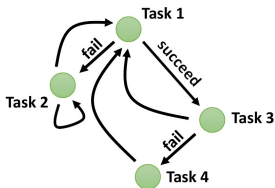
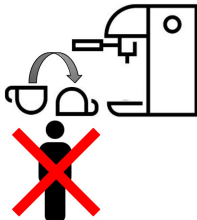


Christiano, Leike, Brown, Martic, Legg, Amodei. [Deep reinforcement learning from human preferences](#). 2017.

How can we learn fully autonomously?



Nagabandi, Konolige, Levine, Kumar. Deep Dynamics Models for Learning Dexterous Manipulation. CoRL 2019.



Task 1: put cup in coffee machine

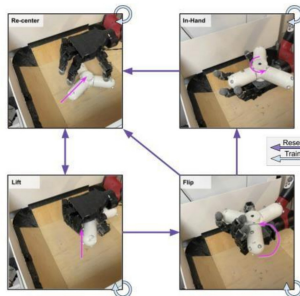
Task 2: pick up cup

Task 3: replace cup

Task 4: clean up spill from cup...

Nagabandi, Konolige, Levine, Kumar. Deep Dynamics Models for Learning Dexterous Manipulation. CoRL 2019.

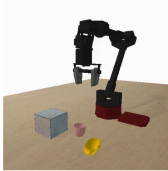
How can we learn fully autonomously?



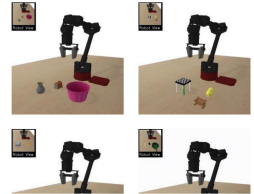
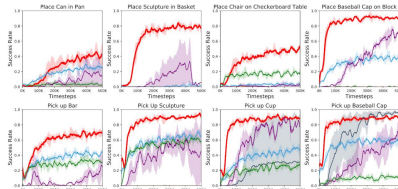
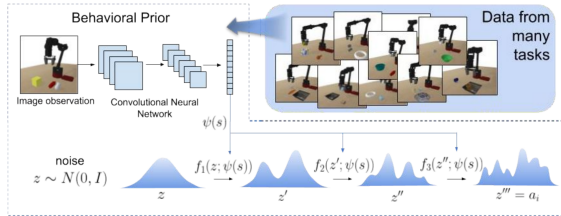
Gupta, Yu, Zhao, Kumar, Rovinsky, Xu, Devlin, Levine. Reset-Free Reinforcement Learning via Multi-Task Learning. 2021.

How to bootstrap exploration from experience?

exploring from scratch



exploring from behavioral prior



Singh*, Hui*, Zhou, Yu, Rhinehart, Levine. Parrot: Data-driven behavioral priors for reinforcement learning. 2020.

This all seems really hard, what's the point?



Why is this interesting?

- ▶ It's exciting to see what solutions intelligent agents come up with
- ▶ Most exciting if they come up with something we don't expect
- ▶ This requires the world they inhabit to admit novel solutions
- ▶ This means that world must be complicated enough!

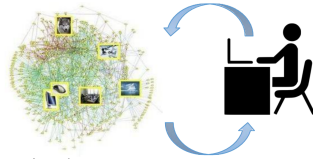
To see interesting emergent behavior, we must train our systems in environments that actually require interesting emergent behavior!

RL in the real world may be difficult, but it is also rewarding.

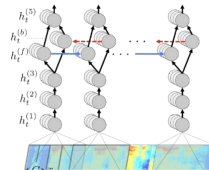
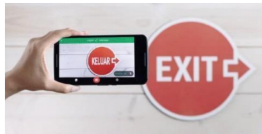
Reinforcement Learning as “**Universal**” **Learning**



The top image shows a long row of server racks in a data center, with many cables plugged into the front. The bottom image is a close-up of a large pile of US dollar bills, mostly \$100 bills, representing wealth or profit.



labeled



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Why do we need **machine learning** anyway?

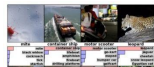
A postulate: We need machine learning for one reason and one reason only — that's to produce adaptable and complex decisions.



Decision: how do I move my joints?



Decision: how do I steer the car?



What is the decision? The image label? **Usually not!**

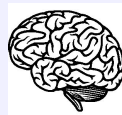
What happens with that label afterwards? *These are **decisions** — they lead to **consequences***

Aside: why do we need **brains** anyway?



**Daniel
Wolpert**

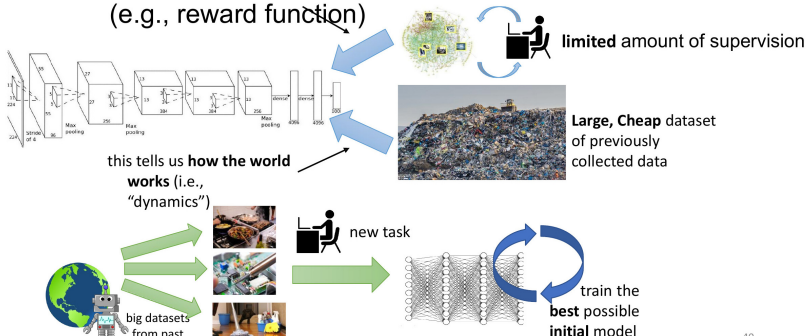
(knows quite a lot
about brains)



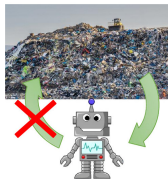
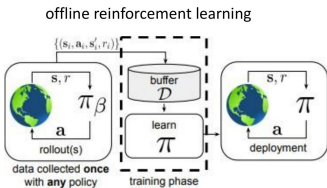
*"We have a brain for one reason and one reason only — that's to produce **adaptable and complex movements**. Movement is the only way we have affecting the world around us... I believe that to understand movement is to understand the whole brain."*

Reinforcement learning as a way to use “cheap” data

- this tells us what the **task** is
(e.g., reward function)



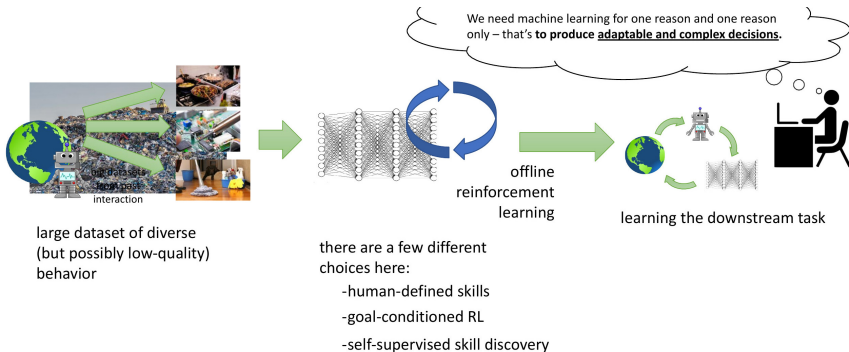
The RL + data problem



with naïve RL, this is a **costly** interactive process if done in the real world!

but self-supervised learning is about using **cheap** data that we already have lying around (in the garbage heap)!

A1



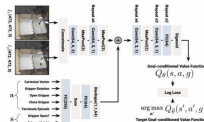
17

Can we learn from offline data without well-defined tasks?

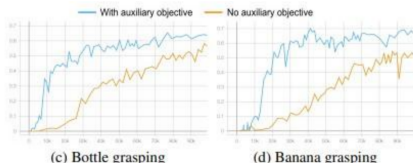
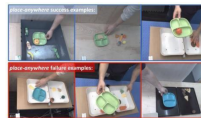
- ▶ No reward function at all, task is defined entirely using a goal image
- ▶ Uses a conservative offline RL method designed for goal-reaching, based on CQL
- ▶ Works very well as an unsupervised pretraining objective



1. Train **goal-conditioned** Q-function with offline RL



2. Finetune with a **task** reward and limited data



Chebatar et al. Actionable Models: Unsupervised Offline Reinforcement Learning of Robotic Skills. 2021.

Can offline RL train large language models?

What we want:



Can you teach me about behavior cloning?



Of course! I'd be happy to explain behavior cloning. To start, could you tell me if you've ever come across the terms "artificial intelligence" or "machine learning" before?



Yes, I've heard those terms, but I'm not exactly sure what they mean.



No problem at all! Let's take it step by step. Have you ever used a computer or a smartphone?

What we get:

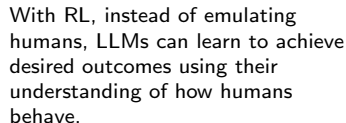
J

Can you explain the AI concept of behavior cloning to me? But first, I want you to ask me questions to figure out what my background in AI is, then tailor your explanation to match my level.



Certainly! Let's start by understanding your background in AI to provide an explanation that matches your level of knowledge. Please answer the following questions:

1. Have you ever studied or worked with artificial intelligence or machine learning concepts before, even at a basic level?
2. Are you familiar with terms like "neural networks," "training data," or "supervised learning"?
3. Have you ever heard of or worked with specific AI tools or frameworks like TensorFlow, PyTorch, or scikit-learn?
4. Are you interested in a high-level overview or a more detailed explanation of behavior cloning?
5. Do you have any specific examples or applications in mind where you'd like to see behavior



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Example: model-based RL for multi-turn dialogue

GPT Agent Dialogue:

H: Hi, I've been hearing about something called "behavior cloning" in AI, but I don't really know what it is. Can you explain it to me?

AI: Certainly! Let's start by understanding your background in AI to provide an explanation that matches your level of knowledge. Please answer the following questions: 1. Have you ever studied or worked with artificial intelligence or machine learning concepts before, even at a basic level? 2. Are you familiar with terms like "neural networks," "training data," or "supervised learning"? 3. Are you interested in a high-level overview or a more detailed explanation of behavior cloning?

H: I have some basic knowledge of artificial intelligence and machine learning, but I wouldn't say I'm an expert. I am familiar with neural networks and supervised learning, but not as much with training data.

AI: Great, thank you for that information. Behavior cloning is a method used in reinforcement learning, which is a subcategory of machine learning. In reinforcement learning, an AI system learns to make optimal decisions by interacting with its environment and receiving feedback in the form of rewards or punishments. Behavior cloning, on the other hand, is a technique where an AI model imitates or "clones" the behavior of an already established agent, instead of learning from scratch through trial and error. This is often used when there is a large amount of data available from an expert or a pre-existing system. Does that make sense so far?

H: Yes, that's starting to make sense. What do you mean by expert?

AI: An expert, in this context, refers to a human or an advanced AI system that has already learned how to solve a particular task or problem. The AI model then observes and imitates this expert's behavior to learn how to perform the task itself. Does that clarify what behavior cloning is?

H: Yes, thank you.

IE-RL Agent Dialogue:

H: Hi there! I've heard about this thing called "behavior cloning" in AI, but I don't really understand what it is. Can you explain it to me?

AI: Of course! I'd be happy to explain behavior cloning. To start, could you tell me if you've ever come across the terms "artificial intelligence" or "machine learning" before?

H: Yes, I've heard those terms, but I'm not exactly sure what they mean.

AI: No problem at all! Let's take it step by step. Have you ever used a computer or a smartphone?

H: Yes, I use my computer and phone all the time.

AI: Great! So, imagine that a computer or a smartphone could learn to do something by watching you do it. That's a bit like what behavior cloning is.

H: Oh, interesting. Can you explain more about how that works?

AI: Sure. Behavior cloning is a type of machine learning algorithm where a computer learns to perform a task by imitating a human's behavior. It does this by closely analyzing and mimicking the actions of a human expert, using a large dataset of examples. This approach is often used in tasks that don't have a clear set of rules or can't be easily solved through traditional programming methods. Does that make sense?

H: Yes, I think so. So how does the computer learn to mimic the behavior?

AI: That's a great question. The computer is fed a large amount of data that shows different examples of the task being performed by a human expert. The computer then uses this data to identify patterns and develop a model of the behavior. This model allows the computer to predict the correct actions to take in a given situation. Over time, the computer can continue to refine its model and improve its performance. Does that help clarify things?

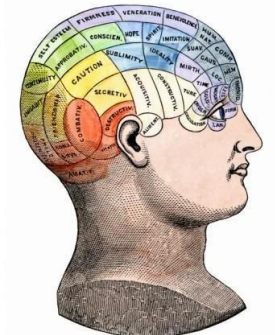
H: That does. Thank you!

Hong, Levine, Dragan. Zero-Shot Goal-Directed Dialogue via RL on Imagined Conversations. 2023.

Back to the **Bigger Picture**

- ▶ Reinforcement learning = can reason about decision making
- ▶ Deep models = allows RL algorithms to learn and represent complex input-output mappings

Deep models are what allow reinforcement learning algorithms to solve complex problems end to end!



How Much Information Does the Machine Need to Predict?

Y LeCun

■ “Pure” Reinforcement Learning (cherry)

- ▶ The machine predicts a scalar reward given once in a while.
- ▶ **A few bits for some samples**

■ Supervised Learning (icing)

- ▶ The machine predicts a category or a few numbers for each input
- ▶ Predicting human-supplied data
- ▶ **10→10,000 bits per sample**

■ Unsupervised/Predictive Learning (cake)

- ▶ The machine predicts any part of its input for any observed part.
- ▶ Predicts future frames in videos
- ▶ **Millions of bits per sample**



■ (Yes, I know, this picture is slightly offensive to RL folks. But I'll make it up)

These slides have been adapted from

- ▶ Chelsea Finn, [CS224R: Deep Reinforcement Learning](#), Stanford
- ▶ Sergey Levine, [CS285: Deep Reinforcement Learning](#), Berkeley